AdifySkills

n8n Automation Mastery
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CLASS NOTES

Module 6 · Production-Grade Engineering

DAY 19 of 25

Self-Hosting n8n: Docker, VPS & Domains

"Apna khud ka n8n server — unlimited workflows."

⚡ AAJ HUM YEH BANAYENGE

Self-Hosted n8n + Uptime Monitor

Aaj VPS par Docker Compose se n8n + Postgres + Caddy (auto HTTPS) chalate hain — ready files neeche hain. Saath mein ek chhota workflow jo har 5 minute aapke n8n ka /healthz check karta hai aur down hone par Telegram alert deta hai.

Docker | Docker Compose | Postgres | Nginx/Caddy

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Self-hosting = unlimited executions, data apne paas, client ko kam cost. Har serious automator ko aana chahiye.

🎯 Aaj ke baad aap yeh kar paoge

01 Cloud vs self-hosted: cost & control

🌍 Kiraya vs EMI.
⚙️ Cloud: easy, limits. Self-host: sasta, full control, maintenance.

02 Docker & Docker Compose basics

🌍 Tiffin box — sab ek dabbe mein.
⚙️ docker-compose.yml: n8n + postgres services, volumes.

03 VPS setup (Hostinger VPS / any Ubuntu server)

🌍 Khaali plot.
⚙️ Ubuntu VPS → SSH → Docker install → firewall.

04 Custom domain + SSL (reverse proxy)

🌍 Ghar ka naam board + taala.
⚙️ DNS A record → Caddy/Nginx reverse proxy → Let's Encrypt SSL.

05 Environment variables & Postgres backend

🌍 Ghar ki settings ek list mein.
⚙️ N8N_HOST, WEBHOOK_URL, N8N_ENCRYPTION_KEY, DB_TYPE=postgresdb.

06 Updates, backups & restore

🌍 Ghar ki duplicate chaabi.
⚙️ pg_dump nightly + encryption key safe; docker compose pull && up -d.

⏱ Class ka plan (2 ghante)

10m	30m	55m	15m	10m

`0:00-0:10` **Recap**

🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

`0:10-0:40` **Concept: Hosting Architecture**

🪄 *Kiraye ke ghar mein maalik ke rules. Apne ghar mein tumhare rules — par safai bhi tumhe karni hai.*

`0:40-1:35` **Live Setup**

Fresh VPS → Docker → n8n + Postgres → domain + SSL

🪄 *Aaj ki kahani yaad hai — 'Kiraye ka flat vs apna ghar'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

`1:35-1:50` **Checklist Review**

Security hardening

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

`1:50-2:00` **Q&A + Homework**

🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

VPS

Cloud mein apna virtual computer (Ubuntu).

🪄 **Analogy:** Kiraye ka flat jahan aap kuch bhi rakh sakte ho

CONCEPT 02

Docker

App + uski saari dependencies ek "container" mein.

🪄 **Analogy:** Tiffin box — sab packed

CONCEPT 03

Reverse proxy + SSL

Domain → n8n, HTTPS ke saath (Caddy/Nginx).

🪄 **Analogy:** Building ka reception

CONCEPT 04

Encryption key

Credentials encrypt karne ki key — kho gayi to credentials gaye.

🪄 **Analogy:** Locker ki master key

🔑 Aaj ka main idea

Self-hosting = n8n ko apne server (VPS) par Docker se chalana. Domain + HTTPS lagao, Postgres database use karo, aur regular backup lo.

- Hostinger VPS par n8n + Postgres
- n8n.meracompany.com HTTPS ke saath
- Har raat database backup Drive par

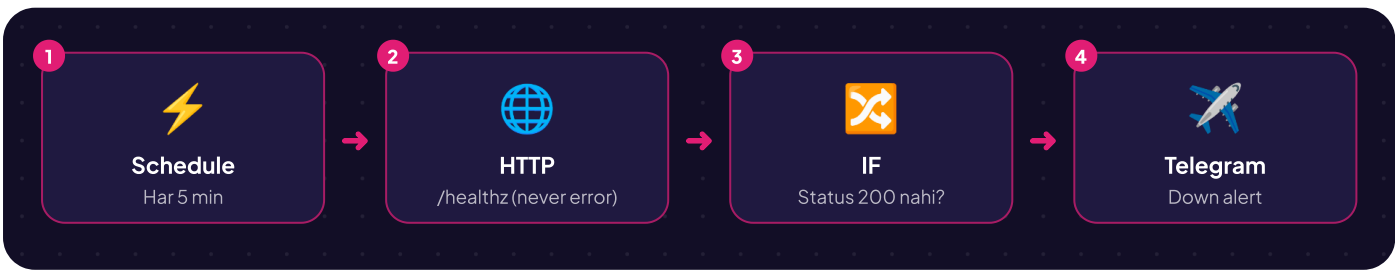
📖 Ek kahani se samjho: Kiraye ka flat vs apna ghar

1. 🏢 **Cloud = kiraye ka flat** = n8n Cloud
2. 🏠 **Apna plot liya** = VPS server
3. 📦 **Ready-made ghar ka dabba** = Docker container
4. 🪧 **Ghar ka address board** = Domain + SSL
5. 🗄️ **Tijori aur backup chaabi** = Postgres + backups

🪄 **Ek line mein: "Kiraye ke ghar mein maalik ke rules. Apne ghar mein tumhare rules — par safai bhi tumhe karni hai."**

🗺️ Aaj ka workflow: Self-Hosted n8n + Uptime Monitor

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Clients jo apna n8n chahte hain (VPS setup + monitoring) — aam taur par **₹8,000 – ₹20,000 + maintenance**.
(Andaaza hai; client aur scope ke hisaab se badalta hai.)

🛠 Step-by-step build — n8n kholo aur saath chalo

1. 🖥️ **SSH**
   VPS mein login, apt update
2. 🖥️ **SSH**
   Docker install: curl -fsSL https://get.docker.com | sh
3. 📄 **Read/Write Files from Disk**
   docker-compose.yml: n8n + postgres
4. 🖥️ **SSH**
   DNS A record → VPS IP; Caddy reverse proxy
5. 🖥️ **SSH**
   docker compose up -d
6. ⚡ **Schedule Trigger**
   Nightly backup workflow

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Telegram bot token (BotFather se)**
- ☐ URL mein apna domain daalo
- ☐ Best practice: yeh monitor kisi DOOSRE n8n / server par chalao (same server down hua to alert kaun bhejega?)

✅ Test kaise karein

1. Galat URL daal ke test karo → alert aana chahiye

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)	⚡ Ab (n8n)
Cloud plan ₹2000+/mahina, limits	VPS ~₹500/mahina, unlimited workflows

⏱ **Kharcha kam, control poora — lekin maintenance tumhari.**

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
services:
  n8n:
    image: docker.n8n.io/n8nio/n8n
    environment:
      - N8N_HOST=n8n.yourbrand.com
      - WEBHOOK_URL=https://n8n.yourbrand.com/
      - DB_TYPE=postgresdb
      - N8N_ENCRYPTION_KEY=${N8N_ENCRYPTION_KEY}
    volumes: [n8n_data:/home/node/.n8n]
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "domain": "n8n.yourbrand.com",
  "vps": "2 vCPU / 4 GB RAM"
}
```

📤 Output 📋 Select karke copy karo

```
{
  "url": "https://n8n.yourbrand.com",
  "ssl": "valid",
  "db": "postgres"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Aaj VPS par Docker Compose se n8n + Postgres + Caddy (auto HTTPS) chalate hain — ready files neeche hain. Saath mein ek chhota workflow jo har 5 minute aapke n8n ka /healthz check karta hai aur down hone par Telegram alert deta hai.

Flow: Schedule (Har 5 min) → HTTP (/healthz (never error)) → IF (Status 200 nahi?) → Telegram (Down alert)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt 📋 Select karke copy karo

```
I'm building the n8n workflow "Self-Hosted n8n + Uptime Monitor" (nodes: Every 5 Minutes, Ping n8n, Down?, Down Alert).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
4 nodes · ready to import
`adifyskills.com/workflows/day-19-uptime-monitor.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/19`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Telegram bot token (BotFather se)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.

⚠️ Yeh galtiyan sabse common hain

- ! WEBHOOK_URL set na karna — webhooks localhost dikhate hain
- ! Encryption key backup na karna
- ! Volume mount bhool jaana — restart par sab data gayab

⭐ Pro tips — time bachane wali tricks

- ★ Hostinger VPS par n8n ka 1-click template bhi milta hai
- ★ Updates se pehle backup + release notes padho
- ★ Firewall: sirf 22, 80, 443 open

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Live: ek fresh Ubuntu VPS par n8n + Postgres Docker Compose ke saath, apne domain par HTTPS.

📝 Homework

Apna self-hosted instance setup karo aur ek workflow export/import karke migrate karo.

- ☐ Homework ho gaya

BEGINNER
Local Docker par n8n chalao ☐

INTERMEDIATE
VPS + domain + SSL ☐

ADVANCED
Automatic backups to S3/Drive ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Monitor ko 2 baar lagataar fail hone par hi alert bhejne do (false alarm kam).

💡 **Hint 1**
Ek baar fail hone par thoda ruk ke dobara check karo.

💡 **Hint 2**
Down? true → Wait 1 minute → Ping dobara → IF phir bhi down → alert.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Webhooks ka public URL batane wala env?

(A) NODE_ENV
(B) WEBHOOK_URL
(C) PORT
(D) TZ

Q2. Credentials encrypt karne wali key?

(A) SSH key
(B) SSL cert
(C) N8N_ENCRYPTION_KEY
(D) API_KEY

Q3. Data restart ke baad bhi rahe — kya chahiye?

(A) Cache
(B) Cookie
(C) Nothing
(D) Docker volume

Q4. HTTPS ke liye?

(A) Reverse proxy + SSL cert
(B) Bada server
(C) Postgres
(D) Redis

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Docker
Software ko ek ready dabbe (container) mein chalane ka tool — har jagah same chalta hai.

VPS
Kiraye ka online computer/server jo 24×7 chalta hai.

SSL
Website ka taala — https wala lock, data safe rehta hai.

Schedule
Fixed time par workflow chalana — roz, har ghante, har Monday.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 20

Scaling, Security & Workflow DevOps

"Ek client se 50 clients tak — bina tootey."

☐ **Day 19 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → B
WEBHOOK_URL

Q2 → C
N8N_ENCRYPTION_KEY

Q3 → D
Docker volume

Q4 → A
Reverse proxy + SSL cert

✅ Challenge ka solution

Down? (true) → Wait 1 min → HTTP Ping n8n (copy) → IF statusCode != 200 → Down Alert.

📚 Official docs

- Hosting n8n ↗
- Docker install ↗
- Environment variables ↗
- Server setups ↗

▶ Videos

- n8n self host docker compose ↗
- n8n hostinger vps install ↗

Doubt poochho
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